Helix Forge Research Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-003394 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000003 (research/literature-notes/2026-07-13/helix-forge-research/metric-contracts/tables/helix-forge-research-metric-contracts-metrics-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	205	0.867	0.871	0.869
slice-02	219	0.884	0.847	0.865
slice-03	197	0.866	0.860	0.863
slice-04	252	0.883	0.873	0.878
slice-05	251	0.865	0.849	0.857